## Exercises

**Exercise 3.1 · Exponential forms.** Write in exponential form: $-2+2i$; $-1-i\sqrt{3}$; $5i$; $\dfrac{1+i}{1-i}$.

**Solution.** $-2+2i = 2\sqrt{2}\,\mathrm{e}^{3i\pi/4}$; $-1-i\sqrt{3} = 2\mathrm{e}^{-2i\pi/3}$; $5i = 5\mathrm{e}^{i\pi/2}$; $\frac{1+i}{1-i} = \frac{\sqrt{2}\mathrm{e}^{i\pi/4}}{\sqrt{2}\mathrm{e}^{-i\pi/4}} = \mathrm{e}^{i\pi/2}$.

**Exercise 3.2 · Two ways of writing the same product.** Let $z_1 = \sqrt{3}+i$ and $z_2 = 1-i$.

1. Write $z_1$, $z_2$ and then $z_1 z_2$ in exponential form.
2. Write $z_1 z_2$ in algebraic form.
3. Deduce the exact values of $\cos\frac{\pi}{12}$ and $\sin\frac{\pi}{12}$.

**Solution.**

1. $z_1 = 2\mathrm{e}^{i\pi/6}$, $z_2 = \sqrt{2}\mathrm{e}^{-i\pi/4}$, $z_1 z_2 = 2\sqrt{2}\mathrm{e}^{-i\pi/12}$.
2. $z_1 z_2 = \sqrt{3} - \sqrt{3}i + i + 1 = \left(\sqrt{3}+1\right) + \left(1-\sqrt{3}\right)i$.
3. Identifying: $2\sqrt{2}\cos\left(-\frac{\pi}{12}\right) = \sqrt{3}+1$ and $2\sqrt{2}\sin\left(-\frac{\pi}{12}\right) = 1-\sqrt{3}$. So $\cos\frac{\pi}{12} = \frac{\sqrt{3}+1}{2\sqrt{2}} = \frac{\sqrt{6}+\sqrt{2}}{4}$ and $\sin\frac{\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}} = \frac{\sqrt{6}-\sqrt{2}}{4}$.

**Exercise 3.3 · Powers.**

1. Compute $(1-i)^8$ and $\left(\sqrt{3}+i\right)^{12}$.
2. Determine the natural numbers $n \geq 1$ for which $\left(\sqrt{3}+i\right)^n$ is real; positive real; purely imaginary. Give the smallest in each case.

**Solution.**

1. $(1-i)^8 = \left(\sqrt{2}\right)^8 \mathrm{e}^{-2i\pi} = 16$ and $\left(\sqrt{3}+i\right)^{12} = 2^{12}\mathrm{e}^{2i\pi} = 4{,}096$.
2. $\left(\sqrt{3}+i\right)^n = 2^n \mathrm{e}^{in\pi/6}$. It is real if and only if $\frac{n\pi}{6} \equiv 0 \pmod{\pi}$, that is $n$ a multiple of 6; the smallest is 6. It is a positive real number if and only if $n$ is a multiple of 12; the smallest is 12. It is purely imaginary if and only if $\frac{n\pi}{6} \equiv \frac{\pi}{2} \pmod{\pi}$, that is $n = 3+6k$; the smallest is 3.

**Exercise 3.4 · De Moivre's formula.** Using de Moivre's formula and the binomial formula, express $\cos 4\theta$ in terms of $\cos\theta$ only, and $\sin 4\theta$ in terms of $\cos\theta$ and $\sin\theta$.

**Solution.** With $c = \cos\theta$ and $s = \sin\theta$: $(c+is)^4 = c^4 + 4ic^3 s - 6c^2 s^2 - 4ics^3 + s^4$. The real part gives $\cos 4\theta = c^4 - 6c^2 s^2 + s^4$; with $s^2 = 1-c^2$, $\cos 4\theta = 8c^4 - 8c^2 + 1$. The imaginary part gives $\sin 4\theta = 4c^3 s - 4cs^3$.

**Exercise 3.5 · Linearising in order to integrate.**

1. Using Euler's formulas, prove that $\sin^3 x = \dfrac{3\sin x - \sin 3x}{4}$.
2. Deduce the value of $\displaystyle\int_0^{\pi} \sin^3 x \, \mathrm{d}x$.

**Solution.**

1. $\sin^3 x = \dfrac{\left(\mathrm{e}^{ix} - \mathrm{e}^{-ix}\right)^3}{(2i)^3} = \dfrac{\mathrm{e}^{3ix} - 3\mathrm{e}^{ix} + 3\mathrm{e}^{-ix} - \mathrm{e}^{-3ix}}{-8i} = \dfrac{2i\sin 3x - 6i\sin x}{-8i} = \dfrac{3\sin x - \sin 3x}{4}$.
2. An antiderivative is $-\frac{3}{4}\cos x + \frac{1}{12}\cos 3x$. So the integral equals $\left(\frac{3}{4} - \frac{1}{12}\right) - \left(-\frac{3}{4} + \frac{1}{12}\right) = \frac{4}{3}$.

**Exercise 3.6 · Modulus and argument of $1 + \mathrm{e}^{i\theta}$.**

1. Prove that for every real number $\theta$, $1 + \mathrm{e}^{i\theta} = 2\cos\frac{\theta}{2}\,\mathrm{e}^{i\theta/2}$.
2. Deduce the modulus and an argument of $1 + \mathrm{e}^{i\theta}$ when $\theta \in (-\pi, \pi)$.
3. Application: modulus and argument of $1 + \mathrm{e}^{2i\pi/3}$, then a check with the algebraic form.

**Solution.**

1. $1 + \mathrm{e}^{i\theta} = \mathrm{e}^{i\theta/2}\left(\mathrm{e}^{-i\theta/2} + \mathrm{e}^{i\theta/2}\right) = 2\cos\frac{\theta}{2}\,\mathrm{e}^{i\theta/2}$ by Euler's formula.
2. For $\theta \in (-\pi, \pi)$, $\frac{\theta}{2} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, so $\cos\frac{\theta}{2} > 0$. The expression is an exponential form: modulus $2\cos\frac{\theta}{2}$, argument $\frac{\theta}{2}$.
3. Modulus $2\cos\frac{\pi}{3} = 1$ and argument $\frac{\pi}{3}$. Indeed $1 + \mathrm{e}^{2i\pi/3} = 1 - \frac{1}{2} + \frac{\sqrt{3}}{2}i = \frac{1}{2} + \frac{\sqrt{3}}{2}i = \mathrm{e}^{i\pi/3}$.

**Exercise 3.7 · A sum of cosines.** Let $\theta$ be a real number that is not a multiple of $2\pi$, and $n$ a natural number.

1. Prove that $\displaystyle\sum_{k=0}^{n} \mathrm{e}^{ik\theta} = \frac{\mathrm{e}^{i(n+1)\theta} - 1}{\mathrm{e}^{i\theta} - 1}$.
2. Using the half-angle method, prove that this sum equals $\mathrm{e}^{in\theta/2}\dfrac{\sin\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}$.
3. Deduce $\displaystyle\sum_{k=0}^{n} \cos(k\theta) = \frac{\cos\frac{n\theta}{2}\,\sin\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}$, then check the formula for $n = 2$ and $\theta = \frac{\pi}{2}$.

**Solution.**

1. It is a geometric sum with common ratio $q = \mathrm{e}^{i\theta} \neq 1$ (since $\theta \not\equiv 0 \pmod{2\pi}$): $\sum_{k=0}^{n} q^k = \frac{q^{n+1}-1}{q-1}$, and $q^k = \mathrm{e}^{ik\theta}$.
2. For every real number $t$, $\mathrm{e}^{it} - 1 = \mathrm{e}^{it/2}\left(\mathrm{e}^{it/2} - \mathrm{e}^{-it/2}\right) = 2i\sin\frac{t}{2}\,\mathrm{e}^{it/2}$. The quotient is

therefore $\dfrac{2i\sin\frac{(n+1)\theta}{2}e^{i(n+1)\theta/2}}{2i\sin\frac{\theta}{2}e^{i\theta/2}} = e^{in\theta/2}\dfrac{\sin\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}$; the denominator is not zero because $\frac{\theta}{2}$ is not a multiple of $\pi$.

3. Take the real part of both sides. For $n = 2$ and $\theta = \frac{\pi}{2}$: $\cos 0 + \cos\frac{\pi}{2} + \cos\pi = 0$, and the formula gives $\frac{\cos\frac{\pi}{2}\sin\frac{3\pi}{4}}{\sin\frac{\pi}{4}} = 0$.

## Problem

**Problem · Superposing two oscillations.** In physics, a simple voltage or sound can be written $t \mapsto A\cos(\omega t + \varphi)$. We show here that a sum $a\cos x + b\sin x$ can be put in this form. We study $g(x) = \sqrt{3}\cos x + \sin x$.

1. Write $\sqrt{3} - i$ in exponential form.
2. Prove that $g(x) = \mathrm{Re}\left(\left(\sqrt{3} - i\right)\mathrm{e}^{ix}\right)$, then that $g(x) = 2\cos\left(x - \frac{\pi}{6}\right)$.
3. Deduce the maximum and the minimum of $g$ on $\mathbb{R}$, and the real numbers in $[0, 2\pi)$ where they are reached.
4. Solve in $[0, 2\pi)$ the equation $g(x) = \sqrt{2}$.
5. Generalise: let $a$ and $b$ be two real numbers, not both zero. Prove that there exist $R > 0$ and a real number $\varphi$ such that, for every real number $x$, $a\cos x + b\sin x = R\cos(x - \varphi)$. Express $R$ in terms of $a$ and $b$.

**Solution · Problem.**

1. $\left|\sqrt{3} - i\right| = 2$, $\cos\theta = \frac{\sqrt{3}}{2}$, $\sin\theta = -\frac{1}{2}$: $\sqrt{3} - i = 2\mathrm{e}^{-i\pi/6}$.
2. $\left(\sqrt{3} - i\right)(\cos x + i\sin x) = \sqrt{3}\cos x + \sin x + i\left(\sqrt{3}\sin x - \cos x\right)$, whose real part is $g(x)$. Moreover $\left(\sqrt{3} - i\right)\mathrm{e}^{ix} = 2\mathrm{e}^{i(x - \pi/6)}$, whose real part is $2\cos\left(x - \frac{\pi}{6}\right)$.
3. The maximum is 2, reached for $x - \frac{\pi}{6} \equiv 0 \pmod{2\pi}$, that is $x = \frac{\pi}{6}$; the minimum is $-2$, reached for $x = \frac{7\pi}{6}$.
4. $\cos\left(x - \frac{\pi}{6}\right) = \frac{\sqrt{2}}{2}$ is equivalent to $x - \frac{\pi}{6} \equiv \frac{\pi}{4}$ or $x - \frac{\pi}{6} \equiv -\frac{\pi}{4} \pmod{2\pi}$, that is $x \equiv \frac{5\pi}{12}$ or $x \equiv -\frac{\pi}{12} \pmod{2\pi}$. In $[0, 2\pi)$: $x = \frac{5\pi}{12}$ or $x = \frac{23\pi}{12}$.
5. The complex number $a - ib$ is nonzero; write it $R\mathrm{e}^{-i\varphi}$ with $R = \sqrt{a^2 + b^2} > 0$. Then $a\cos x + b\sin x = \mathrm{Re}\left((a - ib)\mathrm{e}^{ix}\right) = \mathrm{Re}\left(R\mathrm{e}^{i(x - \varphi)}\right) = R\cos(x - \varphi)$.